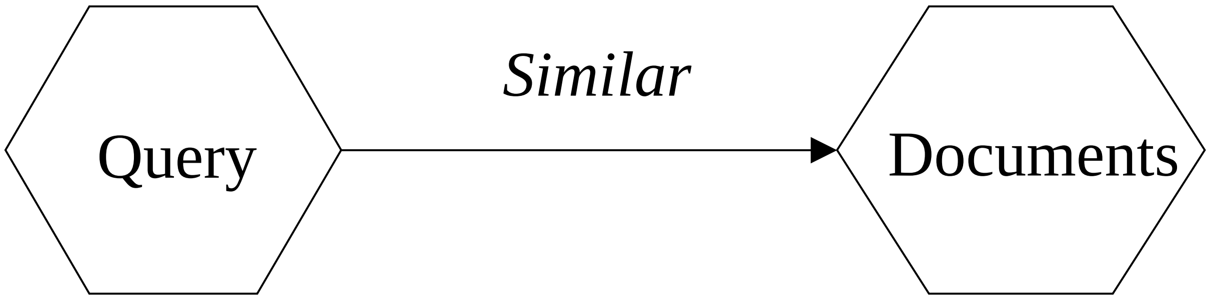
Information Retrieval

Lecture 2

Similarity Ranking Methods

Similarity Ranking Methods

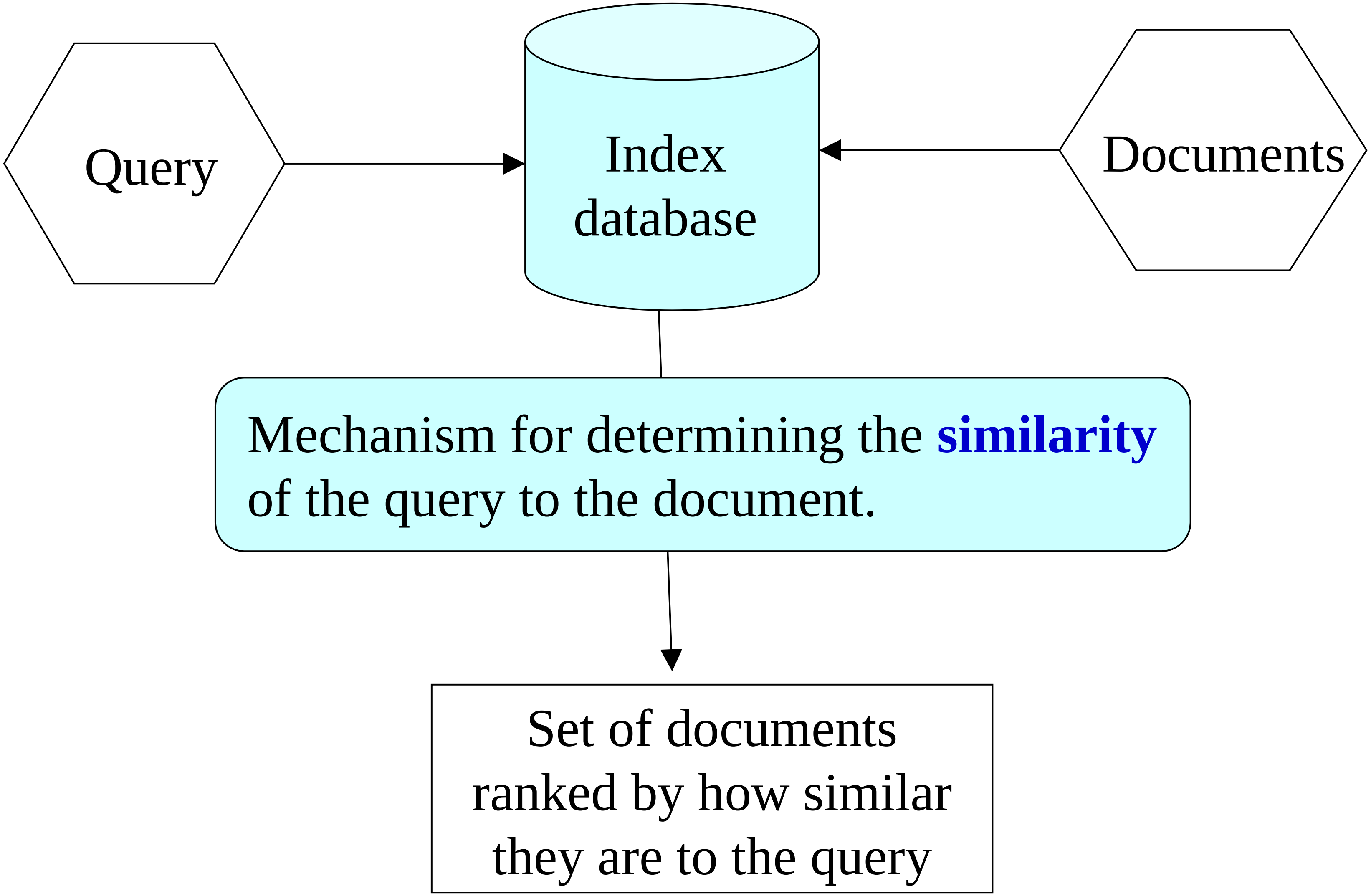
Similarity ranking methods: measure the degree of similarity between a query and a document.



Similar: How similar is a document to a query?

*[Contrast with methods that look for exact **matches** (e.g., Boolean). Those methods assume that a document is either <u>relevant</u> to a query or <u>not relevant</u>.]*

Similarity Ranking Methods: Use of Indexes



Term Similarity: Example

Problem: Given two text documents, how **similar** are they?

A documents can be any length from one word to thousands. The following examples use very short artificial documents.

Example

Here are three documents. How similar are they?

d_1 *ant ant bee*

d_2 *dog bee dog hog dog ant dog*

d_3 *cat gnu dog eel fox*

Term Similarity: Basic Concept

Concept: Two documents are similar if they contain some of the same terms.

Term Vector Space: No Weighting

Term vector space

n-dimensional space, where n is the number of different terms used to index a set of documents (i.e. size of the word list).

Vector

Document i is represented by a vector. Its magnitude in dimension j is t_{ij}, where:

$t_{ij} = 1$ if term j occurs in document i

$t_{ij} = 0$ otherwise

[This is the basic method with no term weighting.]

Basic Method: Term Incidence Matrix (No Weighting)

document	text	terms
d_1	*ant ant bee*	*ant bee*
d_2	*dog bee dog hog dog ant dog*	*ant bee dog hog*
d_3	*cat gnu dog eel fox*	*cat dog eel fox gnu*

	ant	bee	cat	dog	eel	fox	gnu	hog
d_1	1	1						
d_2	1	1		1				1
d_3			1	1	1	1	1	

3 vectors in 8-dimensional term vector space

$t_{ij} = 1$ if document i contains term j and zero otherwise

Similarity between two Documents

Similarity

The **similarity** between two documents, d_1 and d_2, is a function of the **angle** between their **vectors** in the term vector space.

Example: Comparing Documents (No Weighting)

	ant	bee	cat	dog	eel	fox	gnu	hog	*length*
d_1	1	1							2
d_2	1	1		1				1	4
d_3			1	1	1	1	1		5

Example: Comparing Documents

Similarity of documents in example:

	d_1	d_2	d_3
d_1	1	0.71	0
d_2	0.71	1	0.22
d_3	0	0.22	1

Similarity between a Query and a Document

Consider a query as another vector in the term vector space.

The **similarity** between a query, *q*, and a document, *d*, is a function of the **angle** between their **vectors** in the term vector space.

Similarity of Query to Documents (Term Incidence Matrix: no Weighting)

query		
q	*ant dog*	
document	**text**	**terms**
d_1	*ant ant bee*	*ant bee*
d_2	*dog bee dog hog dog ant dog*	*ant bee dog hog*
d_3	*cat gnu dog eel fox*	*cat dog eel fox gnu*

	ant	bee	cat	dog	eel	fox	gnu	hog
q	1			1				
d_1	1	1						
d_2	1	1		1				1
d_3			1	1	1	1	1	

Calculate Ranking

Similarity of query to documents in example:

	d_1	d_2	d_3
q	1/2 0.5	1/√2 0.71	1/√10 0.32

- If the query q is searched against this document set, the ranked results are:
- d_2, d_1, d_3

Simple Uses of Vector Similarity in Information Retrieval

Threshold

For query *q*, retrieve all documents with similarity above a threshold, e.g., similarity > 0.50.

Ranking

For query *q*, return the *n* most similar documents ranked in order of similarity.

[This is the standard practice.]

Term Extending the Basic Concept with Term Weighting

An improved **measure of similarity** might take account of:

(a) Whether the terms are common or unusual

(c) How many times each term appears in a document

d) The lengths of the documents

(e) The place in the document that a term appears

f) Terms that are adjacent to each other (phrases)

Term Vector Space with Weighting

Term vector space

n-dimensional space, where n is the number of different terms used to index a set of documents (i.e. size of the word list).

Vector

Document i is represented by a vector. Its magnitude in dimension j is t_{ij}, where:

$t_{ij} > 0$ if term j occurs in document i

$t_{ij} = 0$ otherwise

t_{ij} is the **weight** of term j in document i.